Gender-Based Price Differences: The Case of Deodorants

Maximiliano Machado*

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Abstract

This article investigates the causes and consequences of gender-based price discrimination— recently defined as the "pink tax". Using data from the deodorant market, I document that female-oriented products are priced higher than male-oriented ones. I estimate a differentiated-product demand and supply model under oligopoly to decompose this price gap. The results show that most of the disparity stems from differences in marginal costs. Counterfactual simulations indicate that enforcing gender price parity does not improve welfare for female consumers as intended. Instead, it leaves their surplus unchanged and leads to welfare gains for male consumers.

Keywords: Economics of Gender, Pink Tax, Demand Estimation, Consumer Welfare

JEL Codes: J16, L13, L66, L81

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1 Introduction

The study of gender disparities has long been central to economics (Goldin, 1990, 2014; Card et al., 2016). More recently, attention has turned to gender-based price differences in retail and service markets, often referred to as the "pink tax"—a term describing the pattern whereby products marketed toward women are priced higher than comparable products marketed toward men, despite having similar physical characteristics. In 2020, the State of New York enacted legislation prohibiting the sale of goods subject to this "implicit tax." The primary aim of the law is to reduce financial barriers faced by women by making female-oriented goods more accessible and addressing gender-based price inequities (State of New York, 2020). Specifically, the law stipulates that sellers:

"*(...) are prohibited from charging a price for two "substantially similar" goods or services, if the goods or services are priced differently based on the gender for whom the goods or services are marketed. (...)*" (State of New York, 2020).[1]

This article examines two central aspects of the so-called pink tax, using the deodorant market as a case study. First, I contribute to the literature by documenting gender-based price disparities and exploring their underlying determinants. Whereas existing work has emphasized differences in own-price elasticities, it has largely overlooked two key factors: marginal costs and within-firm substitution across gender-targeted products. Second, I evaluate the potential effects of legislation such as that enacted in New York, which mandates price parity between goods marketed to different genders.

The deodorant market offers a particularly suitable setting to study gender-based price differences for two main reasons. First, gender switching is limited: products marketed to men are predominantly purchased by male consumers, and likewise for female-targeted products. This segmentation enables clearer identification of consumers' choice sets and facilitates welfare analysis using aggregate data. Second, the gender price gap in deodorants is among the largest observed in consumer goods markets—approximately \$0.45 per ounce (Moshary et al., 2023)—though this gap narrows when controlling for product

[1] "Substantially similar goods" are defined as two goods with little difference in the materials used in production, intended use, function, design, features, and brand. For more information, see https://dos.ny.gov/news/former-governor-cuomo-reminds-new-yorkers-pink-tax-ban-goes-effect-today.

characteristics. This suggests that pricing disparities in this category may be especially responsive to regulatory intervention.

As a first step, I estimate linear regressions to quantify the price premium for female-oriented products relative to similar male-oriented counterparts, controlling for detailed product characteristics such as ingredients and brand ownership. I classify products as female-oriented, male-oriented, or non-gendered—a broader categorization than in prior work. The results indicate that female deodorants are, on average, \$0.06 more expensive per ounce than male deodorants.

Next, I estimate a structural model of demand and supply in the deodorant market. Demand is modelled using a random coefficients framework to capture rich substitution patterns across products, and supply is estimated under the assumption of oligopolistic price competition. This approach allows me to recover product-market-specific marginal costs, which are used in the counterfactual analysis. The estimation results indicate that female-oriented products have a more elastic demand than male-oriented and non-gendered products. Additionally, consumers purchasing male products place greater value on the presence of aluminum, whereas those buying female products assign higher value to alcohol content.

Using the structural model, I conduct counterfactual simulations under three scenarios to isolate the mechanisms contributing to gender-based price disparities: assortment composition within multi-product firms, differences in marginal costs, and differences in price sensitivity across consumer segments. First, I assume that multi-product firms do not internalize substitution across their product lines by modelling each product as if it were owned by a separate single-product firm. Second, I impose that marginal costs are identical for male- and female-oriented products within the same firm when ingredient composition is held constant. Third, I combine both assumptions to assess their joint impact on observed price gaps.

The decomposition reveals that differences in marginal costs account for most of the observed pink tax. When marginal costs are equalized across gender-targeted products, the price gap falls from \$0.06 to \$0.01 per ounce. In contrast, removing within-firm substitution has little impact on the pink tax. Notably, when marginal costs are held constant, male-oriented products become more expensive. This result suggests that the reduced-form evidence is largely driven by higher marginal costs for female-oriented goods.

Finally, I simulate a counterfactual scenario in which firms fully comply with legislation prohibiting gender-based price discrimination for products that are similar in ingredient composition and production cost. The results show that consumers of male-oriented products benefit from the policy, whereas those purchasing female-oriented products do not. This asymmetry arises because the regulation enforces price equalization conditional on marginal costs. Given that demand for male-oriented products tends to be more price elastic, cost parity leads to higher equilibrium prices for these goods. Paradoxically, a regulation intended to improve outcomes for female consumers fails to achieve its primary goal, improving instead the welfare for male consumers.

In terms of producer surplus, I find that the top two firms experience profit gains, whereas the effects for smaller firms is more heterogeneous. The regulation may facilitate an equilibrium that would not arise under unconstrained price competition, as firms would otherwise have incentives to undercut. By imposing price parity across gendered products, the regulation limits firms' ability to deviate in some markets, enabling them to raise prices for certain products without triggering competitive responses.

Related Literature. This article contributes to two main areas of economic research. First, it adds to the literature on differentiated product choice in Industrial Organization. I adopt the approach developed by the seminal work of Berry et al. (1995), as well as the contributions of Nevo (2001), Hausman (1999), Gandhi and Houde (2019) and DellaVigna and Gentzkow (2019) for the selection of instruments to estimate consumer preferences and understand firm behaviour in the deodorant industry. Additionally, I apply the estimation tools developed by Conlon and Gortmaker (2020) to estimate the proposed model.

The second line of research this article contributes to is the literature on gender-based discrimination. This stream of research has examined price and expenditure differences between male and female consumers, beginning with the work of Ayres (1991) and Ayres and Siegelman (1995) on price disparities in the car market, further expanded by Goldberg (1996). Jiang (2024) shows the effects of eliminating gender-based pricing through the abolition of "ladies only" discounts in Japanese movie theatres.

Recent literature has explored price differentiation in retail products, although the evidence does not clearly indicate discrimination against female consumers. Using retail scanner data, Moshary et al. (2023) find that although female prices tend to be higher for some products (bar soap or deodorants), male prices are higher for others (body wash, shampoo, or shaving cream). Similar results, without a strict pattern of higher female prices was found by Guittar et al. (2022) with online prices from major U.S. retailers and Duesterhaus et al. (2011) for personal care products and services.

The recent work by Barnes and Brounstein (2022) sits at the intersection of the two main research streams to which this article contributes. Their findings suggest that female consumers' demand is more elastic than male's, implying that, contrary to the pink tax hypothesis, female consumers pay higher prices because they choose goods with higher marginal costs.

To my knowledge, this is the first article to investigate the effect of gender-parity policies in any retail sector, contributing also to the current literature on pink tax, in line with previous results (Moshary et al., 2023; Barnes and Brounstein, 2022). These results can be thought as inputs for the design of policies intending to reduce gender disparities, illustrating that the main gap is not in prices but in costs.

The literature on gender-based disadvantages is much broader, with significant contributions from labor economics. Research on gender disparities (Goldin, 1990, 2006, 2014; Goldin and Katz, 2007, 2008; Blau and Kahn, 2017; Hegewisch, 2018; Babcock and Laschever, 2003; Card et al., 2016), the distribution of household labor (Goldin, 2014; Sayer, 2010; Kolpashnikova, 2016; Kolpashnikova and Kan, 2021; Gupta et al., 2015), and the quality of employment (Antón et al., 2023; Stier and Yaish, 2014; Lažetić, 2020) are some of the areas where this research has made the most significant advances.

This article proceeds as follows. Section 2 discusses the context of the regulation in New York. Section 3 describes the used data sets, shows basic descriptive statistics and facts about the industry. Section 4 includes reduced-form evidence of gender-based disparities in the deodorant market. Section 5 details the structural model and the estimation procedure. Section 6 presents the results from the estimation of the model. Section 7 presents a price decomposition to identify what is causing the observed pink tax. Section 8 estimates welfare effects of banning the pink tax under firms perfect compliance. It concludes in Section 9 with final comments and future lines of research to pursue.

2 Policy Context

This type of legislation raises several important concerns. First, it presumes the existence of a systematic price premium based on a report by the New York State Government (Bessendorf, 2015), which documents that average prices for female-oriented products exceed those of their male counterparts. However, these comparisons do not control for differences in product characteristics. More recent work by Moshary et al. (2023) finds that, once observable characteristics are accounted for, price differences across gendered products largely disappear, and in some cases, even reverse. As a result, such legislation may have limited effect—or could even reduce female consumers' welfare in some markets.

Second, the legislation implicitly assumes that firms will comply by reducing prices on female-oriented products. However, if a pink tax is present, firms may instead adjust prices on both sides of the market, lowering prices on female products and raising prices on male products, depending on relative demand elasticities. For instance, if male consumers are less price sensitive than female consumers, firms may find it profit-maximizing to increase prices on male products rather than decrease prices on female ones. In such cases, the intended goal of improving affordability for female consumers may not be fully realized.

Third, the legislation assumes that female consumers exclusively purchase female-oriented products, which is not always the case. Whereas gender switching is limited in some categories, such as deodorants, it is more prevalent in others, such as razors.

Consequently, if the regulation results in higher prices for male-oriented products, female consumers who purchase those goods may be adversely affected. Moreover, the policy overlooks the role of non-gendered products, which are also purchased by women. Prices in this category may shift in equilibrium in response to changes in relative elasticities, and increases in the price of non-gendered goods could similarly reduce consumer welfare among women.[2]

Thus, it is not clear that gender-parity legislation will achieve its intended goal of improving welfare for female consumers. Although a pink tax may exist in some markets, such regulations risk harming female consumers who currently purchase lower-priced male-oriented products, as well as those who benefit from relatively inexpensive female-oriented goods. This article provides empirical evidence on the potential welfare consequences of such policies.[3]

3 Data and Descriptive Statistics

I combine data from four sources to study the U.S. deodorant market. The primary dataset is the NielsenIQ Retail Scanner Data (RSD), which provides weekly information on sales and prices from over 90 retail chains, covering approximately 35,000 to 50,000 stores nationwide since 2006. This dataset captures more than 50% of grocery and drugstore sales in the United States. Geographic identifiers are available at the Designated Market Area (DMA) and state levels, allowing me to isolate sales occurring within the State of New York. I use these data both in the reduced-form analysis of pricing effects and in the estimation of the structural model.

The second dataset is the NielsenIQ Consumer Panel Data (CPD), which records the purchases of approximately 40,000 to 60,000 active U.S. households using in-home bar-code scanners. This dataset can be linked to the RSD to incorporate product characteristics and store-level information, also providing household demographics such as size,

[2]Appendix S1 provides a simple model illustrating how relative elasticities shape equilibrium price responses.

[3]Throughout the remainder of the article, I refer to "female" and "male" consumers based on the predominant orientation of the products they purchase, rather than biological sex or gender identity.

age composition, gender, income, and other relevant attributes. I use the CPD to construct a gender classification for products whose descriptions do not clearly identify the intended consumer, categorizing them as female-oriented, male-oriented, or non-gender.[4] Appendix A details the classification methodology to define goods in terms of the targeted gender.

The third dataset comes from Label Insight and provides textual information extracted from product packaging. I use this source to obtain data on product ingredients and to construct ingredient groups based on the first five listed components, which serve as proxies for marginal costs, following the approach of Moshary et al. (2023).[5]

Finally, I incorporate data from the Consumer Promotion Report (CPR), which provides wholesale deodorant prices at the national level. These prices exclude discounts or allowances offered to retailers. Both Label Insight and CPR report Universal Product Codes (UPCs), allowing for reliable matching to products in the NielsenIQ datasets. This data is used to investigate the existence of a pink tax at a wholesale level.

The full dataset covers the period from 2017 to 2023. For the main empirical analysis, however, I restrict attention to the year 2018 and focus on deodorant sales in the State of New York. This restriction is motivated by two considerations. First, I require a pre-policy year to avoid contamination from firms' anticipatory responses to the 2020 gender-pricing legislation. Although the law was implemented in 2020, it was passed in 2019, and public discussion around gender-based price disparities had already gained traction prior to that.[6] Second, including multiple years of monthly-level data would substantially increase the number of markets and computational complexity without necessarily improving parameter identification. Data from the full 2017–2023 period are used descriptively to examine trends in market shares and product entry.

Following evidence of limited price dispersion across stores within the same retail chain

[4]Some brands are straightforward to classify—for example, Old Spice is exclusively marketed to male consumers. Others, such as Dove, produce lines targeting both male and female consumers without always making the distinction explicit.

[5]An ingredient group is defined as the ordered set of the first five ingredients listed on the product label. Order matters, as it reflects the relative concentration of each component. For example, $g_1 = \{\text{Water, Propylene Glycol, Calcium Chloride, Trisiloxane, Parfum Fragrance}\}$ is distinct from $g_2 = \{\text{Water, Propylene Glycol, Calcium Chloride, Parfum Fragrance, Trisiloxane}\}$.

[6]For example, see: https://www.npr.org/transcripts/751440592, accessed on May 5$^{\text{th}}$, 2025.

(DellaVigna and Gentzkow, 2019), I define markets at the DMA–retailer–month level. The time dimension is aggregated to the monthly level to reflect the typical purchase frequency of deodorant products.

Products are defined at the brand–gender–ingredient group level. For example, all deodorants from the brand *Degree* that target female consumers and share ingredient group g_1 are treated as a single product. However, female-oriented *Degree* deodorants with an ingredient group g_2 are treated as distinct products when $g_1 \neq g_2$. This level of granularity accounts for variation in formulation across UPCs within the same brand and gender category.[7]

The deodorant market is highly concentrated, with two dominant firms—Procter & Gamble (P&G) and Unilever—together accounting for over 75% of market share in each year of the sample, as shown in Table 1. A notable distinction between these firms is that P&G has minimal presence in the non-gendered segment, where Unilever holds a dominant position. Each year, the top five firms collectively capture approximately 90% of the market. To simplify the analysis and avoid complications arising from complex ownership structures, I restrict attention to the ten largest manufacturers, which together account for roughly 95% of total market share.

Table 2 summarizes the resulting product set, which includes 270 distinct deodorant products manufactured by the ten largest firms. Male-oriented products account for the largest share of total sales during the period (40.3%), followed by female-oriented products (35.8%). In terms of chemical composition, non-gendered products differ markedly from gendered ones, exhibiting a significantly higher concentration of aluminum and organic ingredients. These compositional differences may signal higher product quality, which could help explain observed price variation across categories.

Whereas average prices per item are similar across male- and female-oriented products, substantial differences emerge when comparing unit prices measured per ounce, as shown in Table 3. Female-oriented products exhibit higher unit prices, primarily due to smaller package sizes. The average size of a male deodorant is 3.21 ounces, compared to 2.72

[7]Imputing ingredient groups as the modal group within a brand may obscure meaningful variation in formulation and product positioning.

ounces for female products. A similar pattern holds for wholesale prices. In terms of quantities sold, female-oriented products have higher average monthly sales than both male- and non-gendered products.

Table 1: Yearly Market Shares Across Gender.

Firms	2017	2018	2019	2020	2021	2022
(a) All Products						
P&G	31.5	32.5	32.2	33.8	37.3	38.6
Unilever	44.9	45.5	47.2	47.1	44.7	44.9
Colgate Palmolive	7.7	7.5	7.4	6.5	5.8	5.0
Church and Dwight	4.1	3.7	3.2	2.3	2.3	2.2
(b) Female Products						
P&G	43.6	44.2	42.7	41.7	42.6	42.4
Unilever	40.4	41.0	43.9	45.8	46.5	48.3
Colgate Palmolive	5.0	4.4	4.2	3.4	2.9	2.2
Church and Dwight	0.1	0.1	0.1	0.1	0.0	0.1
(c) Male Products						
P&G	40.9	41.5	40.8	42.0	47.1	48.8
Unilever	36.6	39.0	39.9	39.6	34.6	34.0
Colgate Palmolive	11.2	10.1	10.1	9.2	8.5	7.4
Church and Dwight	0.5	0.3	0.3	0.3	0.2	0.2
(d) Non-gendered Products						
P&G	0.0	0.0	0.0	0.3	4.1	5.7
Unilever	64.3	63.4	65.1	64.6	63.6	61.9
Colgate Palmolive	6.0	7.3	7.6	6.2	5.8	5.9
Church and Dwight	15.2	14.5	12.3	12.3	11.9	11.9

Table 2: Number of Products, Firms, Sales and Product Composition.

	Female	Male	Non-gender
# of products	103	167	100
# of firms	8	8	9
% of total sales	35.8%	40.3%	23.8%
% of products with ethanol	48.5%	52.0%	44.0%
% of products with aluminum	17.5%	12.0%	25.0%
% of products with organic ingredients	1.0%	0.0%	5.0%

Table 3: Descriptive Statistics for Retail Prices, Sales and Wholesale Prices.

	Female		Male		Non-gendered	
	Mean	S.D.	Mean	S.D.	Mean	S.D.
price	5.05	2.05	4.96	1.78	4.61	2.24
price/Oz	2.11	1.25	1.75	1.01	1.75	1.01
Size	2.72	2.37	3.21	11.55	2.85	0.96
Sales (monthly)	161.21	739.23	127.87	473.52	135.36	747.01
Wholesale price	3.67	1.39	3.48	1.46	3.15	1.34
Wholesale price/Oz	1.51	0.79	1.22	0.68	1.18	0.59

3.1 Data Limitations and Considerations

A key limitation of the dataset is that it does not capture product heterogeneity beyond observable characteristics such as ingredient composition, brand, and targeted gender. Products in the health and beauty category often include unobserved features—such as fragrance, moisturizing properties, or packaging aesthetics—that may play an important role in consumer preferences.

To address this limitation, I make two assumptions. First, I assume that unobserved product characteristics are constant across products within a brand–gender–ingredient group. For example, all *Dove* deodorants targeting female consumers and sharing the same first five ingredients are assumed to have identical unobserved attributes (e.g., fragrance). Second, I assume that unobserved characteristics do not systematically affect marginal costs across gendered products. That is, the cost of incorporating fragrance or other non-observable features is assumed to be similar for male- and female-oriented products.

Whereas other product categories, such as razors, may permit relaxation of these assumptions, they present their own empirical challenges. In particular, gender switching is more prevalent in those markets, making it difficult to identify the buyer's gender and, consequently, to estimate gender-specific welfare effects. The deodorant market offers a cleaner setting in this respect.

4 Descriptive Evidence on the Pink Tax

This section presents reduced-form evidence on gender-based price differences in the deodorant market. I quantify the price gap between female- and male-oriented products using linear regression. The empirical strategy differs from that of Moshary et al. (2023) in two key respects. First, I allow for a third category of non-gendered products, whereas their methodology classifies all goods as either female- or male-oriented. Second, I estimate the gender price gap separately for wholesale and retail prices. This distinction enables me to examine whether gender-based price differences arise at the manufacturing level, the retail level, or both.

Specifically, I estimate the following regression:

$$y_{imt} = \beta_{male} male_{it} + \beta_{non} non_{it} + X_i' \alpha + \gamma_{mt} + \gamma_f + \epsilon_{imt} \tag{1}$$

The dependent variable y_{imt} is either the retail price per ounce (p_{imt}) or the wholesale price per ounce (w_{it}). Note that wholesale prices are sourced from the Consumer Promotion Report (CPR) and are recorded at the national level, without variation across markets or retailers.

In this specification, the coefficients β_{male} and β_{non} capture the average price differences between male- and non-gendered products relative to female-oriented products, respectively. The vector X_i includes controls for ingredient-based production costs, such as fixed effects for the group of the first five listed ingredients, and indicator variables for the presence of alcohol, aluminum, and organic components. I include market-by-time fixed effects γ_{mt} to account for local demand and competitive conditions, and manufacturer fixed effects γ_f to control for brand-level pricing strategies.

Table 4 reports the regression results. Two main findings emerge. First, from panel (a), consistent with Moshary et al. (2023), estimates that do not control for ingredient composition or brand ownership reveal a sizable gender price gap of approximately \$0.36 per ounce. Second, this gap is substantially reduced once observable product characteristics are included, falling to approximately \$0.06 per ounce. Given an average product size of 3 ounces, this implies a remaining price gap of roughly \$0.2 per product.

These results suggest that female-oriented products are indeed more expensive, but much of the observed price gap can be explained by differences in ingredient composition and brand ownership. In addition, non-gendered products tend to be priced higher than male-oriented products but lower than female-oriented ones, positioning them in between the two categories in terms of unit pricing.

The second key finding is suggests that the pink tax does not originate solely at the retail level, as can be seen in panel (b). Wholesale prices for female-oriented products are also higher than those for male-oriented products, even after controlling for observable characteristics. This indicates that gender-based price differences are present upstream, at the manufacturer level. Notably, the pink tax legislation in New York applies to both retailers and wholesalers, making this distinction relevant for evaluating its scope and potential effects.

Table 4: Regression Results from Pink Tax Quantification.

	(1)	(2)	(3)	(4)	(5)	(6)
	(a) Retail prices			(b) Wholesale prices		
β_{male}	-0.363***	-0.279***	-0.064***	-0.326***	-0.210***	-0.088***
	(0.007)	(0.007)	(0.005)	(0.031)	(0.031)	(0.021)
β_{non}	-0.371***	-0.086***	-0.013***	-0.228***	-0.030	-0.023
	(0.008)	(0.009)	(0.005)	(0.035)	(0.044)	(0.036)
N	117,857	117,857	117,857	3,927	3,927	3,920
R^2	0.049	0.126	0.782	0.246	0.337	0.818
Market	Y	Y	Y	Y	Y	Y
Manuf.	N	Y	Y	N	Y	Y
Ing.	N	N	Y	N	N	Y

Clustered s.e. at the market level in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5 Model

As documented in the previous section, a persistent price gap exists across gender-targeted products, even after controlling for observable characteristics, market conditions, and firm fixed effects. Similar patterns have been reported in other markets (Duesterhaus et al., 2011; Moshary et al., 2023; Guittar et al., 2022). However, the extent to which this gap is driven by differences in price sensitivity, marginal costs, or firms' internal substitution incentives remains unclear.

To disentangle these mechanisms, I estimate a structural model of demand and supply in the deodorant market in the State of New York. The model allows for rich substitution patterns across products and enables the recovery of product-specific marginal costs. I use the estimated model to decompose the sources of observed price differences and to simulate counterfactual policies aimed at eliminating gender-based price discrimination.

5.1 Consumer Demand

Consumer demand is defined over quantities measured in ounces, and all prices are expressed in per-ounce terms. Each consumer i in market t derives utility from purchasing product j according to the following specification:

$$\begin{aligned} U_{ijt} = \alpha p_{jt} &+ \beta_f fem_j + \beta_m male_j + \beta_e alc_j + \beta_a alum_j + \beta_s size_j \\ &\beta_{f\times e} fem_j \times alc_j + \beta_{m\times e} mal_j \times alc_j + \beta_{f\times a} fem_j \times alum_j + \beta_{m\times a} mal_j \times alum_j + \\ &\sigma_0 v_i + \sigma_{price} v_i p_{jt} + \gamma_{B(j)} \times \gamma_{I(j)} + \gamma_t + \xi_{jt} + \epsilon_{ijt} \end{aligned} \tag{2}$$

In the utility specification, fem_j and $male_j$ are indicator variables denoting whether product j is marketed toward female or male consumers, respectively. The baseline category consists of non-gendered products. The variables $alum_j$ and alc_j indicate whether the product contains aluminum or alcohol in its formulation.[8] To capture preference heterogeneity by consumer gender, I interact these ingredient variables with indicators for the targeted gender.

The variable $size_j$ represents the average package size of product j in logarithmic form. Whereas all prices and quantities are expressed per ounce, including size in the utility function allows for consumer preferences over packaging format. For example, conditional on the same unit price, consumers may prefer smaller packages that are more convenient to carry.

[8] I exclude the indicator for organic ingredients, as it is infrequently observed in the defined product set, as discussed in Section 3.

To allow for heterogeneous price sensitivity across consumers, I include a random coefficient v_i on the price variable, which have a standard normal distribution. The specification also incorporates fixed effects for brand ($\gamma_{B(j)}$) interacted with ingredient-group based on the first five listed components ($\gamma_{I(j)}$). I include market fixed effects (γ_t), where a market is defined as a retailer–DMA–month. Unobserved product-market characteristics, denoted by ξ_{jt}, capture demand shocks that are observable to firms but not to the econometrician and may be correlated with prices. The idiosyncratic error term ε_{ijt} follows a Type I Extreme Value distribution, yielding the standard logit structure for the choice probabilities.

I do not include consumer demographics in the demand specification, as the defining product characteristics are themselves functions of the targeted consumer base. That is, firms design products to appeal to particular demographic segments, implicitly embedding demographic preferences into observed characteristics. Also, focusing only on New York does not provide enough variation in demographics (especially income) across markets.

Following Berry et al. (1995) with the notation in Conlon and Gortmaker (2020), the utility is composed by a product-market mean-utility term (δ_{jt}), a consumer-specific utility term (μ_{ijt}) and the idiosyncratic error term ϵ_{ijt}. In each market, consumers will choose j from among a set of available products $\mathcal{J}_t = \{0, 1, ..., J_t\}$, being $j = 0$ the outside option of not buying any product. Consumer chooses j that satisfies $U_{ijt}(\delta_{jt}, \mu_{ijt}) > U_{ikt}(\delta_{kt}, \mu_{ikt})$ for any $k \in \mathcal{J}_t, k \neq j$.

The assumption that ϵ_{ijt} is distributed type I extreme value allows to formulate market shares as:

$$s_{ijt}(\delta_{ijt}) = \int \mathbb{1}(U_{ijt} > U_{ikt}) d\mu_{it} \quad \forall k \neq j \tag{3}$$

$$s_{ijt}(\delta_{ijt}) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{\sum_k \exp(\delta_{kt} + \mu_{ikt})} d\mu_{it} \tag{4}$$

The set of parameters to be estimated can be divided in two. First, θ_1 includes the demand parameters $\{\beta,\ \alpha\}$, which enter the utility linearly. Second, θ_2 includes the random coefficients parameters $\{\sigma_0, \sigma_{price}\}$, which enter the demand non-linearly.

Parameters are estimated using Generalized Method of Moments (GMM), with the `pyblp` framework by Conlon and Gortmaker (2020).

To address the endogeneity of prices, I use the variant of BLP-style instruments proposed by Gandhi and Houde (2019), which rely on the idea that markups are increasing in product differentiation. Specifically, for each product characteristic k, I construct instruments based on the average absolute difference between product j and other products in the same market t:

$$d_{jt}^{k} = \frac{1}{|\mathcal{J}_t \setminus j|} \sum_{j' \in \mathcal{J}_t,\ j' \neq j} \left| x_{jt}^{k} - x_{j't}^{k} \right|.$$

These instruments are constructed for three product characteristics: size ($size_j$), alcohol content (alc_j), and aluminum content ($alum_j$). The intuition is that greater differentiation in observable characteristics reduces substitution pressure, which relaxes competitive constraints and allows for higher prices. Because these measures are based only on exogenous product attributes, they are plausibly uncorrelated with unobserved demand shocks ξ_{jt}.

5.2 Supply Side

On the supply side, I model a set of multi-product firms $\mathcal{F}$, where each firm $f \in \mathcal{F}$ produces and sells a portfolio of products $\mathcal{J}_f^t$ in market t. Firms compete by setting prices for each of their products, taking as given the prices of rival firms.

For each product $j \in \mathcal{J}_f^t$, firm f chooses its price to maximize profits in market t, solving the following problem:

$$\max_{p_{jt}} \pi(p_{1t}, ..., p_{Jt}) = \sum_{k \in \mathcal{J}_f^t} p_{kt} Q_{kt}(p_{1t}, ..., p_{Jt}) - C(Q_{kt}(p_{1t}, ..., p_{Jt})) \tag{5}$$

where $Q_{kt}(p_{1t}, ..., p_{Jt}) = s_{kt}(p_{1t}, ..., p_{Jt}) M_t$, with M_t the size of market t, and s_{kt} the market share defined in equation (4). Given that markets are defined at the retailer-DMA level, using simply DMA population will not bring an adequate market size. In order to solve this issue I follow Conlon et al. (2021) and use the purchases of dairy products at each store to estimate market sizes (see Appendix S.2. for details).

Taking first-order conditions from equation (5) and assuming constant marginal costs of production:

$$Q_{jt} + \sum_{k \in \mathcal{J}_f} \frac{dQ_{kt}}{dp_{jt}}(p_{kt} - mc_{kt}) = 0 \tag{6}$$

The second term in equation (6) includes both, how changes of p_{jt} affects its own demand, and how it affects the demand of the rest of the goods produced by firm f. First-order conditions across all firms and products can be collected in matrix form:

$$\mathbf{Q}_t + \Omega\Delta_d[\mathbf{p}_t - \mathbf{mc}_t] = 0 \tag{7}$$

With $\mathbf{Q}_t$, $\mathbf{p}_t$ and $\mathbf{mc}_t$ $\mathcal{J}_t \times 1$ vectors of sales, prices and marginal costs for market t, with $\mathcal{J}_t$ the number of products in market t. Ω is the $\mathcal{J}_t \times \mathcal{J}_t$ ownership matrix, such that $\Omega_{k,l} = 1$ if both k and l are produced by the same firm. Δ_d is a $\mathcal{J}_t \times \mathcal{J}_t$ matrix of demand price-sensitivities, such that its (k, l) element is $\frac{dQ_{kt}}{dp_{lt}}$. Once the demand-side parameters are estimated, marginal costs can be recovered by solving the linear equation system in equation (7).

6 Demand Estimation Results

Demand parameter estimates are reported in Table 5. The results indicate a lower willingness to pay for male-oriented products compared to both female-oriented and non-gendered alternatives. The median own-price elasticity across all products is approximately -3.88, consistent with estimates from related categories—for example, Gordon et al. (2013) reports a median elasticity of 3.03. Figure 1 presents the distribution of elasticities by product gender category. Female-oriented products exhibit the highest elasticity in absolute value, with a median of -4.21, compared to -3.75 for male-oriented and -3.80 for non-gendered products.

To assess the robustness of the counterfactual simulations presented in the following sections, I also estimate a more parsimonious version of the demand model. These estimates are reported in Appendix B and yield qualitatively similar results.

The estimates reveal that non-gendered products carry, on average, higher mean utility than both male- and female-oriented goods, with the difference being smaller relative to female products. This finding may be driven by asymmetric cross-gender purchasing patterns: female consumers are more likely to purchase non-gendered products than male consumers. Using data from the Consumer Panel Dataset (CPD), I show in Appendix A that 40.8% of deodorant purchases by female consumers are for non-gendered products, compared to 32.1% for male consumers.

The demand estimates also reveal systematic heterogeneity in preferences. Consumers, on average, tend to prefer smaller package sizes, even when prices are normalized per ounce. The value placed on aluminum in male-oriented products is higher than the value for this ingredient in female-oriented products. The reverse is observed for alcohol.

In the next section, I use the estimated demand model to decompose the observed pink tax into three components: differences in price sensitivity, marginal costs, and substitution patterns across products within the same firm portfolio.

Table 5: Demand Parameter Estimates

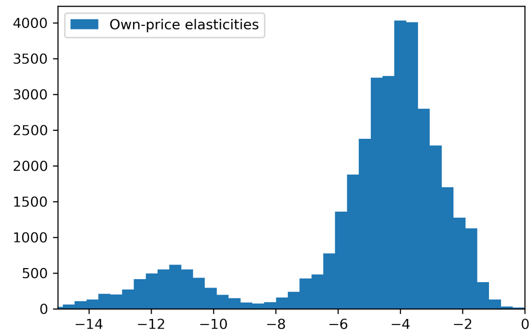
	β	σ
Constant	-	1.0112***
	-	(0.0001)
Price	-2.4586***	0.0001
	(0.4302)	(1.5420)
Female	-0.4469***	-
	(0.0692)	-
Male	-1.0566***	-
	(0.1087)	-
Aluminum	0.1919**	-
	(0.0900)	-
Alcohol	-0.2637**	-
	(0.0883)	-
Size	-1.2552**	-
	(3403)	-
Female × Aluminum	-0.6404***	-
	(0.0382)	-
Male × Aluminum	0.7389***	-
	(0.1190)	-
Female × Alcohol	0.6502***	-
	(0.0461)	-
Male × Alcohol	0.3132***	-
	(0.0685)	-
Median elasticity	-3.885	
Number of Firms	10	
Number of markets	848	
N	117,624	

Clustered s.e. at the market level in parentheses.

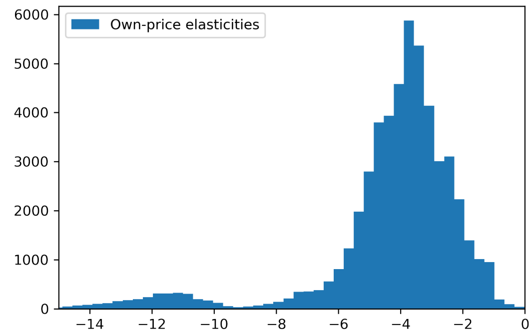
${}^{*}p < 0.1$, ${}^{**}p < 0.05$, ${}^{***}p < 0.01$.

Figure 1: Elasticities by Targeted Gender.

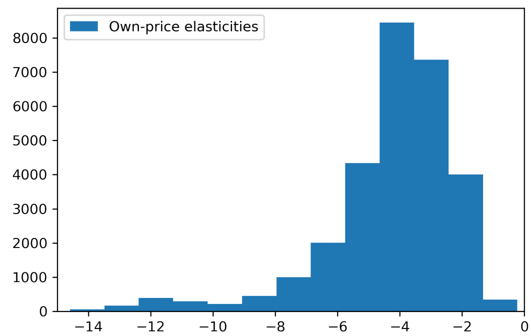
(a) Female-oriented products



(b) Male-oriented products.



(c) Non-gender products.



7 Decomposing the Pink Tax

As shown in Section 4, there exists a persistent price gap between female- and male-oriented products. To better understand the underlying mechanisms driving this disparity, I conduct a decomposition exercise based on the structural model. The goal is to isolate the contribution of three key forces: differences in consumer price sensitivity, differences in marginal costs, and the internalization of substitution effects by multi-product firms.

This decomposition is grounded in firm f's pricing problem, as defined in equation (5), and its corresponding first-order conditions (FOCs), which characterize optimal pricing behavior.

$$p_j = \underbrace{-\left(\frac{\partial Q_j}{\partial p_j}\right)^{-1} Q_j}_{1} + \underbrace{c_j}_{2} \underbrace{- \left(\frac{\partial Q_j}{\partial p_j}\right)^{-1} \left[\sum_{k \in \mathbf{J}_f, k \neq j} (p_k - c_k) \left(\frac{\partial Q_k}{\partial p_j}\right)\right]}_{3} \tag{8}$$

Condition (8) highlights three determinants of equilibrium prices: (i) the own-price sensitivity of demand, (ii) marginal costs of production, and (iii) the substitution (or cannibalization) effect arising from within-firm product portfolios. The magnitude of the substitution effect depends both on the closeness of product k to product j in the consumer's choice set and on the number of products offered by firm f.

If these three components are identical across products with the same composition but different gender orientation, no systematic price gap should arise. To determine the extent to which each factor contributes to the observed pink tax, I simulate a series of counterfactual pricing equilibria that sequentially eliminate gender-based differences in each of the three components.

First, I eliminate the third component—within-firm substitution—by assuming that firms do not internalize cannibalization effects when setting prices. In this scenario, each product is priced as if it were produced by a single-product firm. This removes the incentive to adjust prices based on substitution patterns within a firm's portfolio.

This counterfactual is particularly relevant given that multi-product firms often offer a larger set of female-oriented products than male-oriented ones (i.e., $|\mathcal{J}_f^{female}| > |\mathcal{J}_f^{male}|$). Moreover, substitution among female products may be stronger, potentially leading to higher equilibrium prices due to greater cannibalization pressure. By removing this channel, I isolate the impact of other forces—specifically, differences in marginal costs and price sensitivity—on the observed price gap. If a pink tax persists under this scenario, it must be driven by one or both of these remaining components.

The second counterfactual eliminates gender-based differences in marginal costs for products with identical chemical composition. Specifically, for any firm f that sells two products j and j' such that $X_j = X_{j'}$ —where X_j denotes the group of the first five ingredients along with indicators for aluminum and alcohol content— and either $female_j = male_{j'} = 1$ or $male_j = female_{j'} = 1$, I set their marginal costs equal to the average[9]:

$$\bar{c}_j = \bar{c}_{j'} = \bar{c} = \frac{c_j + c_{j'}}{2} \tag{9}$$

This counterfactual controls for compositional cost differences across gendered products, allowing me to isolate the role of price sensitivity and within-firm substitution in driving the observed pink tax. If a price gap persists under this scenario, it must arise from these remaining channels.

The third counterfactual combines the two previous scenarios, eliminating both within-firm substitution and marginal cost differences across gendered products. Under this setting, any remaining price gap between female- and male-oriented products must be attributed solely to differences in consumer price sensitivity.

For each counterfactual, I solve for equilibrium prices and re-estimate equation (1) to quantify the implied pink tax under the adjusted pricing environment. These counterfactual estimates are then compared to the observed pink tax from Section 4. Results from this decomposition exercise are reported in Table 6.

As shown in Table 6, removing the substitution component from firms' pricing problem has no meaningful effect on the observed pink tax. This suggests that internal cannibalization effects—arising from differences in product portfolio size or substitution intensity across gendered goods—do not drive the price gap. Instead, the disparity must result from either cost differences or heterogeneous price sensitivities.

In the second counterfactual, where firms face identical marginal costs for female- and male-oriented products with the same composition, the pink tax is substantially reduced but not eliminated. The remaining gap is approximately \$0.01 per ounce, indicating that cost asymmetries explain the majority—but not all—of the observed price differential.

Finally, in the third scenario, where both within-firm substitution and cost differences are removed, the results are virtually unchanged from the second case. This implies that the residual pink tax is driven entirely by differences in consumer price sensitivity across

[9] The results are robust to using alternative aggregation methods such as the median, minimum, or maximum of c_j and $c_{j'}$. This is an example with two matched goods, but it can be extended to firms having M male goods matched to F female goods, with $\bar{c} = \frac{1}{F+M} \sum_{k \in F \cup M} c_k$

gendered products.

Taken together, these results indicate that the primary driver of the pink tax is the difference in marginal costs between female- and male-oriented products. A smaller portion of the price gap is attributable to differences in consumer price sensitivity. Specifically, even in a setting where marginal costs are identical across gendered products, female-oriented goods remain more expensive by approximately \$0.014 per ounce, reflecting residual differences in demand elasticities.

One potential concern with the cost-based decomposition is that, although I control for product composition using ingredient groups, this approach may not fully capture all variation in marginal costs. For example, male-oriented products may contain a higher concentration of aluminum than female-oriented products, even when both share the same ingredient group classification. If such within-group variation is systematically related to gender targeting, then some cost asymmetries may remain unaccounted for, potentially biasing the attribution of the pink tax across components.

I re-estimate the pink tax with the inclusion of the marginal costs as a regressor, and check price changes under the decomposition exercise. The equation estimated is:

$$y_{imt} = \beta_{male} male_{it} + \beta_{non} non_{it} + mc_{jt} + X_i'\alpha + \gamma_m + \gamma_t + \gamma_f + \epsilon_{imt} \tag{10}$$

Here, mc_{jt} denotes the marginal cost of product j in market t, recovered from the system of first-order conditions in equation (7). As reported in Column 1 of Table 7, once marginal costs are included as a control, the estimated pink tax disappears: the unconditional price gap of approximately \$0.06 per ounce falls to zero.

This result underscores a key insight from the decomposition exercise: conditional on observable characteristics, female-oriented products are more expensive not due to markup differences, but because they incur higher marginal costs. In Column 3 of the same table, I show that under the counterfactual scenario where marginal costs are equalized across gender-targeted products, the price gap reverses and male-oriented products become more expensive. This reversal further supports the conclusion that cost asymmetries are the primary driver of the observed gender price gap in this market.

These results have a clear economic interpretation. As shown in Section 6, male-oriented products face more inelastic demand. Therefore, when marginal costs are held constant across gendered products, standard pricing logic implies that male products should command higher prices.

Taken together, these findings suggest that in the market for deodorants, there is no evidence of a gender-based gap. Although female-oriented products tend to be more expensive, this is fully accounted for by higher production costs. This result is consistent with the findings of Barnes and Brounstein (2022), who document both higher marginal costs and more elastic demand for female products in other consumer categories.

An additional insight from the counterfactual analysis is that non-gendered products emerge as the most expensive category, both with and without controlling for marginal costs. Under this gender classification, female-oriented products become the least expensive in the market.

As a robustness check, I verify that these findings are not sensitive to the demand specification. In Appendix B, I report results from a more parsimonious demand model, which yields similar estimates for the pink tax and confirms the robustness of the decomposition magnitudes.

In light of these findings, it is important to evaluate the potential effects of policies aimed at eliminating gender-based price discrimination. In the next section, I examine the welfare implications of imposing a pricing constraint that requires firms to set equal prices for male- and female-oriented products with identical ingredient composition and similar marginal costs.[10]

[10]This policy counterfactual is designed to closely mirror the regulatory framework implemented in New York. Matching on ingredients and manufacturer ensures compliance with the "substantially similar goods" condition, whereas conditioning on similar marginal costs reflects the interpretation that price differences may be justified only when cost differentials exist.

Table 6: Pink Tax Decomposition

	(1) Baseline	(2) No substitution	(3) No cost gap	(4) Both
β_{male}	-0.0636***	-0.0611***	-0.0123*	-0.0140***
	(0.0046)	(0.0045)	(0.0066)	(0.0046)
β_{non}	-0.0140**	-0.0135**	0.0336***	0.0324***
	(0.0055)	(0.0054)	(0.0079)	(0.0055)
Market	Y	Y	Y	Y
Manuf.	Y	Y	Y	Y
Ing.	Y	Y	Y	Y
N	117,920	117,920	117,920	117920
R^2	0.776	0.769	0.723	0.764

Column (1) in this Table is the analogous to column (3) in Table 4, though here I drop some observations that have extreme results from the demand estimation (e.g., negative marginal costs, extremely high own-demand elasticities). However, the effect does not seem to dramatically change. Columns (2), (3) and (4) show price differences from prices in scenarios 1, 2 and 3 respectively. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Pink Tax Decomposition With Costs.

	(1) Baseline	(2) No substitution	(3) No cost gap	(4) Both
β_{male}	-0.0001	-0.0002***	0.0312***	0.0349***
	(0.0002)	(0.0001)	(0.0023)	(0.0024)
β_{non}	-0.0012***	-0.0002**	0.0396***	0.0442***
	(0.0002)	(0.0001)	(0.0028)	(0.0029)
mc	1.005***	1.006***	0.696***	0.676***
	(0.0001)	(0.00003)	(0.0015)	(0.0016)
Market	Y	Y	Y	Y
Manuf.	Y	Y	Y	Y
Ing.	Y	Y	Y	Y
N	117,920	117,920	117,920	117920
R^2	0.910	0.889	0.867	0.845

Column (1) in this Table is the analogous to column (3) in Table 4, though here I drop some observations that have extreme results from the demand estimation (e.g., negative marginal costs, extremely high own-demand elasticities). However, the effect does not seem to dramatically change. Columns (2), (3) and (4) show price differences from prices in scenarios 1, 2 and 3 respectively. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

8 Counterfactual Simulation of Price Regulation

In 2020, the state of New York enacted legislation prohibiting sellers from charging different prices solely based on the targeted gender. Evaluating the policy using only factual data (in an event-study framework) is challenging, as male prices rose relative to female prices not only in New York but also in other East Coast states (see Appendix S.3). Part of this increase is attributable to sharp rises in the prices of common inputs such as aluminum and alcohol during this period (see Appendix S.4). Moreover, enforcement of the policy remains unclear, with no official record of government actions taken in response to gender-based price discrimination.

It is not obvious that such legislation would strictly benefit female consumers, for two main reasons. First, in some markets, male-targeted products are already priced higher, so the effect may result in price reductions for male rather than female products. Second, the policy implicitly assumes that female consumers purchase only female-targeted products, whereas their welfare also depends on non-gendered goods (and in some cases also on male-targeted products). If prices for non-gendered goods rise in the new equilibrium, both female and male could be adversely affected.

Here I propose a counterfactual experiment to evaluate the potential welfare effects of such type of regulations, under perfect compliance of firms. In this problem, firms solve equation (5), under the constraint that goods with the same ingredient composition must have the same price, if there are no significant differences in production cost.[11] The problem will then be:

$$\max_{p_j} \pi(p_{1t}, ..., p_{Jt}) = \sum_{k \in \mathcal{J}_f^t} p_{kt} Q_{kt}(p_{1t}, ..., p_{Jt}) - C(Q_{kt}(p_{1t}, ..., p_{Jt})) \tag{11}$$

$$\text{s.t. } p_{ht} = p_{gt} \text{ if } \begin{cases} X_h = X_g \text{ and } female_h = male_g = 1,\ |c_{ht} - c_{gt}| < \epsilon, \text{ or} \\ X_h = X_g \text{ and } male_h = female_g = 1,\ |c_{ht} - c_{gt}| < \epsilon \end{cases}$$

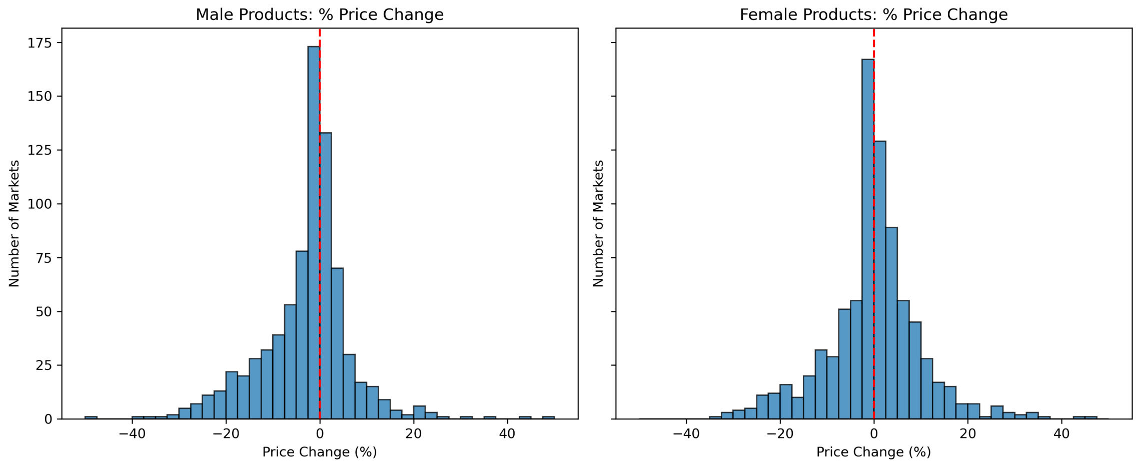
With X_j containing the group of first five ingredients and dummy variables for content of alcohol and aluminum. I choose a threshold $\epsilon = 0.5$ to highlight that the regulation only affects pairs of goods that have similar production costs. In Appendix S.6. I show the results persist under a threshold of $\epsilon = 0.25$. The threshold is used to study a case as close as possible to the regulation in New York, where prices must match unless there are cost differences.

Before analyzing welfare variations, it is important to understand how do counter-

[11] The legislation enacted in New York only allows for price differences when costs justify it.

factual prices behave.[12] Comparing the median prices for female- and male-oriented products with and without legislation, I observe than female-oriented products' prices increase in 50.6% of the markets. On the other hand, male-oriented products increase in 36.3% of the markets. Figure 2 shows histograms of the change in median female- and male-oriented prices.

Figure 2: Distribution of Changes in Median Prices



Around 90% of the price increments are done by large firms, whereas those are responsible also for 68% of price decreases. In Appendix S.5. I show that price increments are more common for female-oriented goods in markets where more products are offered. On the other hand, male-oriented prices tend to increase in markets where more firms compete.

Using these new prices, I compute consumer and firm surplus in each market, compared to the baseline case with no regulation. Figure 3 shows the distribution of welfare changes across markets for female and male consumers, as well as producer surplus for large and small firms.[13]

Both female and male consumers are better-off in more than half of the markets. In Table 8 I show the total welfare change (summed across all markets) and the number of markets where gains are registered for each of type of consumer and firm, where it

[12]For the following analysis I eliminate products which counterfactual prices per ounce are above $10, which corresponds to around 2.8% of the sample. This is done because in some markets the convergence in the algorithm is not obtained and prices en up being substantially large in the new equilibrium.

[13]I define large firms to be P&G and Unilever, which dominate around 75% of the market, being the remaining manufacturers defined as small firms.

is shown that female consumers have an increase of 1.6% and male consumers one of in 1.8%. This is driven by a mixture of effects. First, the policy conditions on costs, with prices being matched for male and female products that have a similar marginal costs. As shown in Section 4, male products have a more inelastic demand, hence, at the same marginal costs, male products are expected to have higher prices. The legislation would likely decrease prices for male products.

Second, the gains for female consumers are mostly explained by non-gender products. In Figure 4 I show welfare changes by considering only gendered products, i.e., female consumers buy only female-oriented goods, and male consumers buy only male-oriented goods. The overall welfare effects are displayed in column (3) of Table 8. Notice that welfare from female-oriented goods is unchanged, whereas welfare from male-oriented goods increases by 1.0%. This reflects the results in terms of price increments, with female-oriented products increasing in 50% of the markets, whereas male-oriented products only in 36%.

This means that the main objective of the regulation is not achieved. Those consumers buying predominantly female-oriented products are not better off. In contrary, consumers buying male-oriented and non-gendered deodorants are the ones with higher welfare.

This result goes in line with the findings from the previous section, in the sense that, conditioning on costs, male products are expected to be priced higher due to a more inelastic demand. Even though this change in welfare is not large enough overall, it creates welfare losses that go up to 50% in some markets, for both female and male consumers. Hence, despite not disrupting the overall welfare, the heterogeneous effects on consumers should be a concern.

An interesting result emerges when looking at firms' side, with large firms (Unilever and P&G) earning higher profits in most markets, whereas the effect for small firms is more heterogeneous, though with more markets where profits decrease. Overall, large firms' profits increase by 52.9%, finding larger profits in almost all markets. The interpretation here relies on this legislation allowing for an equilibrium that is not supported under pure price competition. There is a set of prices $\mathcal{P}_t = \{p_{0t}, p_{1t}, ..., p_{Nt}\}$ for all N

goods in market t, that improves firms profits but is not stable under price competition, but it is under another type of competition (e.g., a price agreement). Now, the legislation may be allowing for such an equilibrium forcing firm to raise some prices.

This mechanism contributes to a reduction in competitive intensity. Although firms may wish to compete à la Bertrand—driving prices closer to marginal cost—the regulation restricts their ability to do so. Lowering the price of one product now requires lowering the price of others, increasing the cost of undercutting. As a result, firms face fewer profitable deviations from high-price strategies, weakening price competition and pushing market outcomes closer to those observed under monopoly pricing.

Table 8: Summary of Welfare Changes.

	(1) ΔWelfare	(2) % markets w/ΔWelfare < 0	(3) ΔWelfare (gendered)
Male	1.8%	35.7%	1.0%
Female	1.6%	37.4%	0.0%
Large Firms	52.9%	13.3%	-
Small Firms	13.9%	59.0%	-

Column (1) reports changes from considering female (male) consumers have non-gendered and female (male) products in their choice set. Column (3) reports changes by eliminating non-gendered products from choice sets.

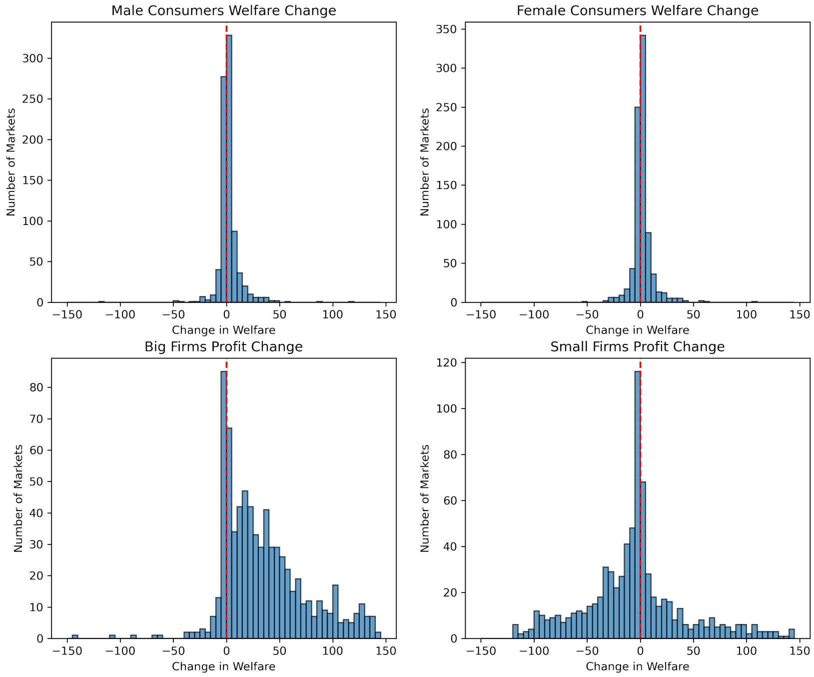


Figure 3: Distribution of Welfare Changes by Gender and Type of Firm.

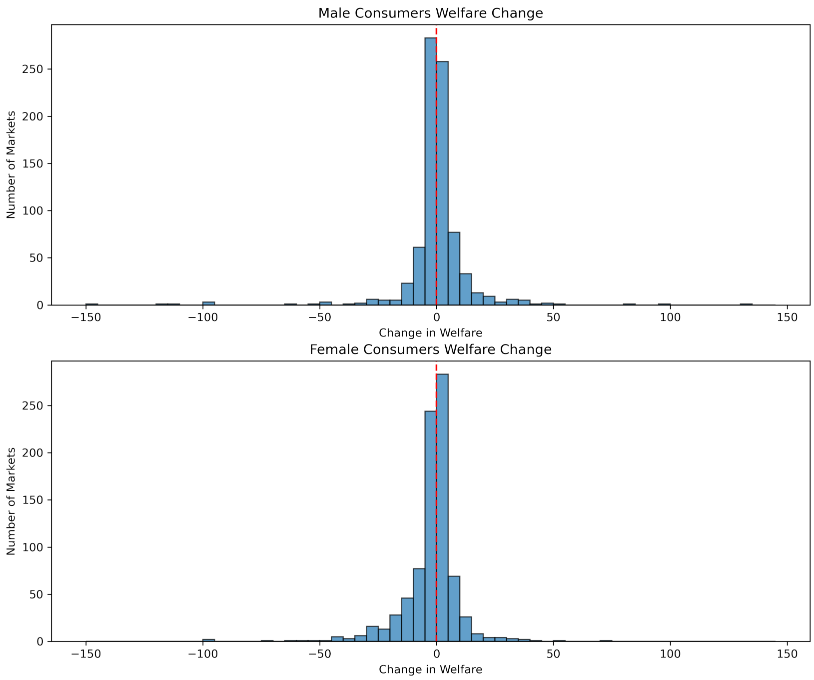


Figure 4: Distribution of Welfare Changes from Gendered Products.

9 Conclusion

This article evaluates the existence and drivers of gender-based price disparities in the deodorant market and assesses the effects of regulatory interventions. Using rich scanner and panel data, combined with a structural demand model, I decompose the so-called "pink tax" and find that, not controlling for marginal costs of production leads to the incorrect conclusion that female products tend to be more expensive than male ones. Once marginal costs are controlled for, the pink tax is inexistent, highlighting higher costs of production for female products, even when composition and ownership is the same across goods.

Counterfactual simulations show that imposing gender price parity, even when conditioned on cost similarity, can produce unintended welfare effects. Although both female and male consumers may benefit in aggregate, female consumers who primarily purchase female-oriented products are not better-off. On the supply side, large multi-product firms tend to benefit from the regulation through constrained price adjustments, suggesting that such policies can induce equilibria that are not supported in pure price competition. These findings underscore the importance of targeting the source of disparities —namely, cost differences— rather than enforcing uniform retail pricing.

One important point to mention here is that this model does not allow neither for endogenous product choice nor firms exit. Under this type of legislation, once two goods must match prices, the firm could opt to remove one of them to avoid the price change of the other, if this would disrupt profits less. This would have two side effects: (i) firms could have higher profits than the estimated here, and (ii) consumer could lose welfare due to reduction in variety. If it allows for exit and entry, some of the small firms that see their profits decrease would probably choose to exit. This, similarly to the case of endogenous product choice, would generate a reduction product variety, and increase market power for large firms. Both these elements would contribute to a reduction in consumer welfare, unless entry of new firms compensates.

Future work should explore whether similar dynamics hold in other product categories and examine long-run firm responses such as product repositioning, entry, or exit.

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Appendix

A. Gender Classification and Purchases

Gender Classification

I classify goods in terms of gender using three alternatives. First, I classify goods based on their brands. Some brands sell only goods that target a particular gender, e.g., Old Spice targets male consumers or Secret targets female consumers. I recover a list of the most popular brands from online sellers and use Chat GPT 4.0 to get a classification of the targeted gender for each of the brands observed in NielsenIQ data. Then, I match the classified brands with the NielsenIQ data. Brands that sell goods that can target more than one gender (e.g., Degree sells both male- and female products) are not classified in this step. Additionally to using AI I also classify as female-oriented all brands having the words “LADY”, “WOMAN”, or “WOMEN”, whereas brands containing “MAN” or “MEN” are classified as male-oriented. In Table A1 I outline which brands I define as female- and male-oriented, and then every UPC in these brands will be classified into the respective gender.

An issue with this classification method is that some brands produce goods targeting multiple genders, and the brand name alone does not allow for a clear classification. For those cases, in second step, I use the description provided in the CPR dataset to generate a classification using Chat GPT 4.0, and then match these goods to the NielsenIQ dataset.[14]

Finally, for those goods that cannot be classified using neither of the previous two methods (i.e., brands where not found in online sellers nor in CPR data) I use the methodology employed by Barnes and Brounstein (2022), using the CPD from Nielsen IQ. Considering only single-person households (27% of households in the US), I compute the share of purchases of each product done by female and male consumers. Products purchased more than 50% of times by female (male) consumers are classified as female (male).

[14]To comply with NielsenIQ data policies regarding the use of AI, no NielsenIQ data was fed into any generative AI tool. The data used was external and later matched to NielsenIQ products.

After these steps I obtain a gendered classification for each of the UPCs considered into either female- or male, and the unclassified goods will be coded as non-gendered.

Female brands	Male Brands
CRYSTAL	AOS ART OF SPORT
DRY IDEA	AXE
ELIZABETH ARDEN	BEVEL
LOVE BEAUTY AND PLANET	BRUT
MEGABABE	DR. SQUATCH
SECRET	DOVE MEN
SEVENTH GENERATION	DUKE CANNON SUPPLY CO.
SOFT & DRI	EVERY MAN JACK
LADY SPEED STICK	GILLETTE
	MENNEN
	OBSESSION FOR MEN
	OLD SPICE
	RIGHT GUARD
	SPEED STICK
	CK
	CURVE

Table A1: Gendered Brands.

Gendered Purchases

Table A2 presents the share of purchases from each type of good for female and male consumers in the CSD. To avoid having consumers buying goods to individuals of other gender I consider only single-person households form this dataset, which corresponds to around 25% of the total sample. I use the same gender classification as outlined before.

	female	male	non-gendered
Female	0.478	0.113	0.408
Male	0.065	0.613	0.321

Table A2: Gendered Purchases by Consumer Gender.

B. Robustness Checks

Here I report the results from using a more parsimonious demand model, which I will refer as the "alternative" demand specification. Results include the parameters estimated in Table B1, the decomposition of the pink tax in Tables B2 and B3, and the counterfactual welfare changes from a regulation banning gender-based price discrimination in Table B4. The plots with distribution of changes across markets can be found in Appendix S.6.

Table B1: Demand Parameter Estimates

	β	σ
Constant	-	-5.3055
	-	(11.9628)
Price	-2.9041***	0.6477**
	(0.4096)	(0.3028)
Female	-0.0152	-
	(0.0119)	-
Male	-0.4695***	-
	(0.0167)	-
Median elasticity	-4.291	
Number of Firms	10	
Number of markets	848	
N	117,624	

Clustered s.e. at the market level in parentheses.
${}^{*}p < 0.1$, ${}^{**}p < 0.05$, ${}^{***}p < 0.01$.

Table B2: Pink Tax Decomposition

	(1) Baseline	(2) No substitution	(3) No cost gap	(4) Both
male_prod	-0.0644***	-0.0644***	-0.0133***	-0.0100*
	(0.00461)	(0.00460)	(0.00394)	(0.00393)
nongender_prod	-0.0133*	-0.0123*	0.0315***	0.0358***
	(0.00550)	(0.00549)	(0.00470)	(0.00468)
Market	Y	Y	Y	Y
Manuf.	Y	Y	Y	Y
Ing.	Y	Y	Y	Y
N	117,920	117,920	117,920	117920
R^2	0.776	0.769	0.723	0.764

Column (1) in this Table is the analogous to column (3) in Table 4, though here I drop some observations that have extreme results from the demand estimation (e.g., negative marginal costs, extremely high own-demand elasticities). However, the effect does not seem to dramatically change. Columns (2), (3) and (4) show price differences from prices in scenarios 1, 2 and 3 respectively. ${}^{*}p < 0.1$, ${}^{**}p < 0.05$, ${}^{***}p < 0.01$.

Table B3: Pink Tax Decomposition With Costs.

	(1) Baseline	(2) No substitution	(3) No cost gap	(4) Both
male_prod	-0.00113**	-0.00114***	0.0307***	0.0329***
	(0.000430)	(0.000322)	(0.00232)	(0.00239)
nongender_prod	-0.00118*	-0.000102	0.0400***	0.0440***
	(0.000513)	(0.000383)	(0.00276)	(0.00285)
estimated_costs	1.035***	1.035***	0.719***	0.703***
	(0.000284)	(0.000212)	(0.00153)	(0.00157)
Market	Y	Y	Y	Y
Manuf.	Y	Y	Y	Y
Ing.	Y	Y	Y	Y
N	117,920	117,920	117,920	117920
R^2	0.998	0.999	0.943	0.938

Column (1) in this Table is the analogous to column (3) in Table 4, though here I drop some observations that have extreme results from the demand estimation (e.g., negative marginal costs, extremely high own-demand elasticities). However, the effect does not seem to dramatically change. Columns (2), (3) and (4) show price differences from prices in scenarios 1, 2 and 3 respectively. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	(1) ΔWelfare	(2) % markets w/ΔWelfare < 0	(3) ΔWelfare (gendered)
Men	2.6%	37.6%	0.4%
Women	2.2%	36.9%	-1.2%
Large Firms	70.1%	14.5%	-
Small Firms	23.2%	60.5%	-

Column (1) reports changes from considering women (men) have non-gendered and female (male) products in their choice set. Column (3) reports changes by eliminating non-gendered products from choice sets.

Table B4: Summary of Welfare Changes With Threshold of \$0.5 (Alt. Specification).